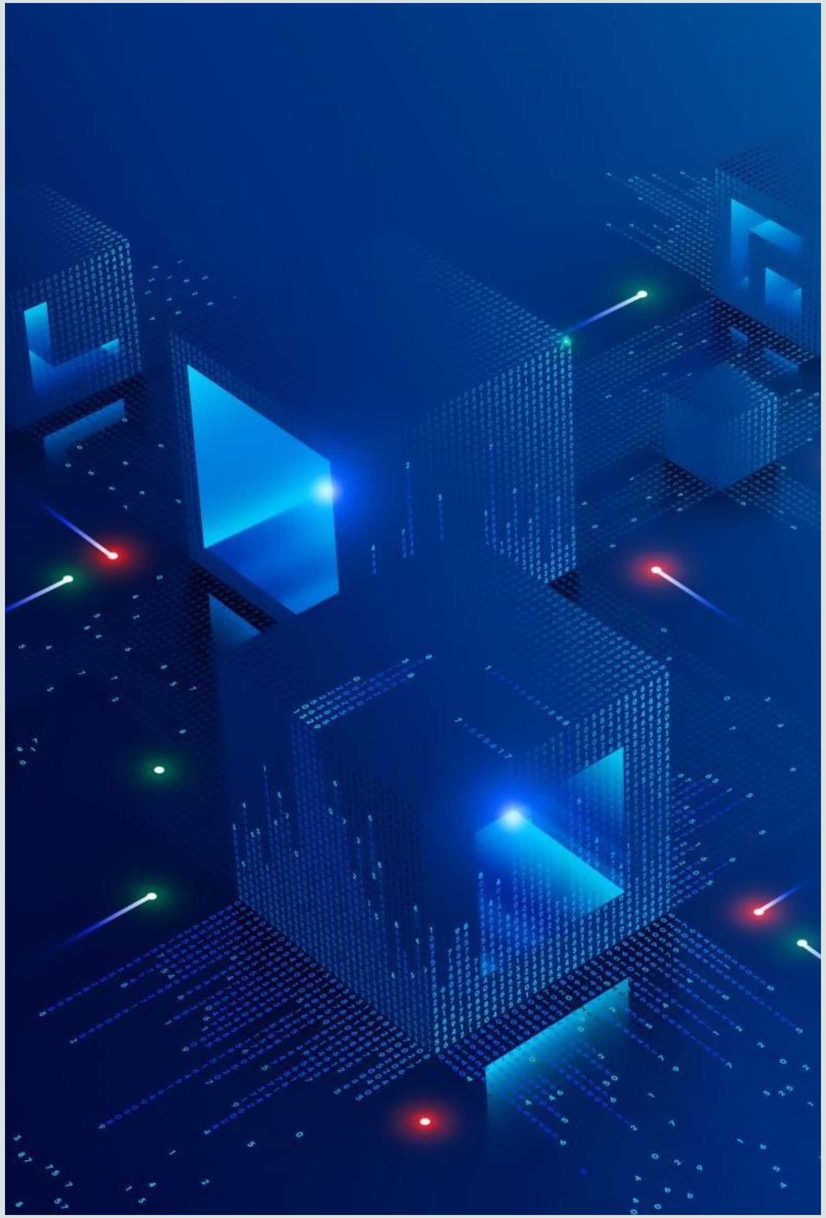

Quantum formalism to describe binocular rivalry (Efstratios Manousakis)

By

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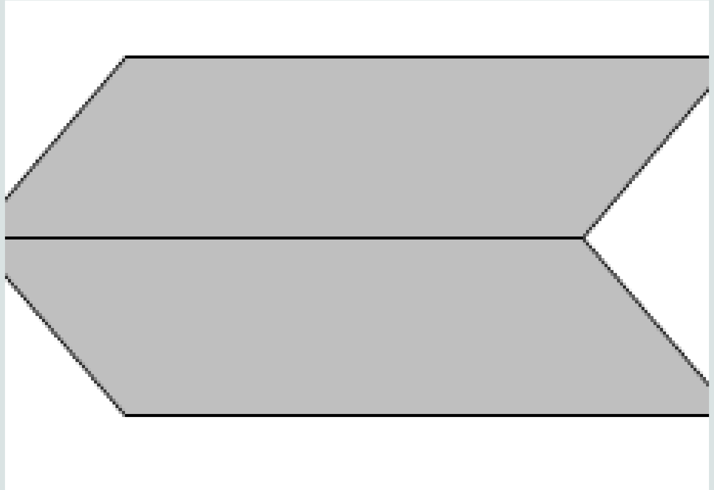
CONTENT

Introduction

- **Binocular rivalry** is a visual phenomenon wherein one experiences alternating perceptions due to the occurrence of different stimuli presented to the corresponding retinal regions of the two eyes and their competition for perceptual dominance.
- The process of perception, a formalism is sought to mathematically describe the **subjective or abstract/mental** process of perception.
- It is shown that the formalism of **orthodox quantum theory** of measurement, where the observer plays a key role, is a broader mathematical foundation which can be adopted to describe the dynamics of the subjective experience.
- The mathematical formalism describes the **psychophysical dynamics** of the subjective or cognitive experience as communicated to us by the subject. Subsequently, the formalism is used to describe simple perception processes and, in particular, to describe the probability distribution of dominance duration obtained from the testimony of subjects experiencing binocular rivalry.
- Using this theory and parameters based on known values of **neuronal oscillation frequencies and firing rates**, the calculated probability distribution of dominance duration of rival states in binocular rivalry under various conditions is found to be in good agreement with available experimental data. This theory naturally explains an observed marked increase in dominance duration in binocular rivalry upon periodic interruption of stimulus and yields testable predictions for the distribution of perceptual alteration in time.

General formulation



- **Potentiality and actuality:**(Dirac notation)In order to describe the dynamics of such perceptual or mental events we will use the notion of **potential consciousness** and we will show that such a tool is very useful in describing the evolution and distribution of dominance durations in binocular rivalry.The set of all possible potential events |ii forms a complete basis in a **Hilbert space** describing all potential outcomes.

$$|v\rangle = c_1|1\rangle + c_2|2\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \qquad (1)$$

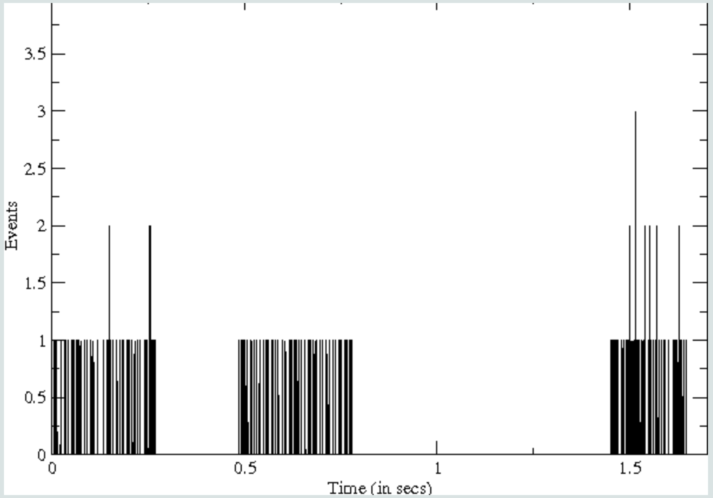
- **Operation of Consciousness and observation:**The previous example tells us, that the stimuli have in some sense an “unfathomable” existence, or they exist as potentialities, namely, we cannot talk about them directly but only of how they are perceived, i.e., after they have been operated upon by the process of **conscious perception**.This operation of consciousness called perception or measurement causes an actual experience and an actual conscious event.Eg:Dinner Table Conversation
- **Evolution of potential consciousness:(**Schrodinger Equation).When a perceptual event is observed, where a wave-function “collapse” occurs through the process of measurement, it actualizes the corresponding **neural correlate of consciousness** (NCC) brain state.

APPLICATION TO BINOCULAR RIVALRY

- **Time evolution of potential consciousness**:Through this operation of consciousness, the conscious percept (or symbol) acts and activates the corresponding neural correlates of consciousness (NCC) in the nervous system.**State vector formulation** and **Density matrix formulation**. If after the measurement is “no” the state of potential consciousness is represented by a density matrix given by ˆ1 − Qˆ.

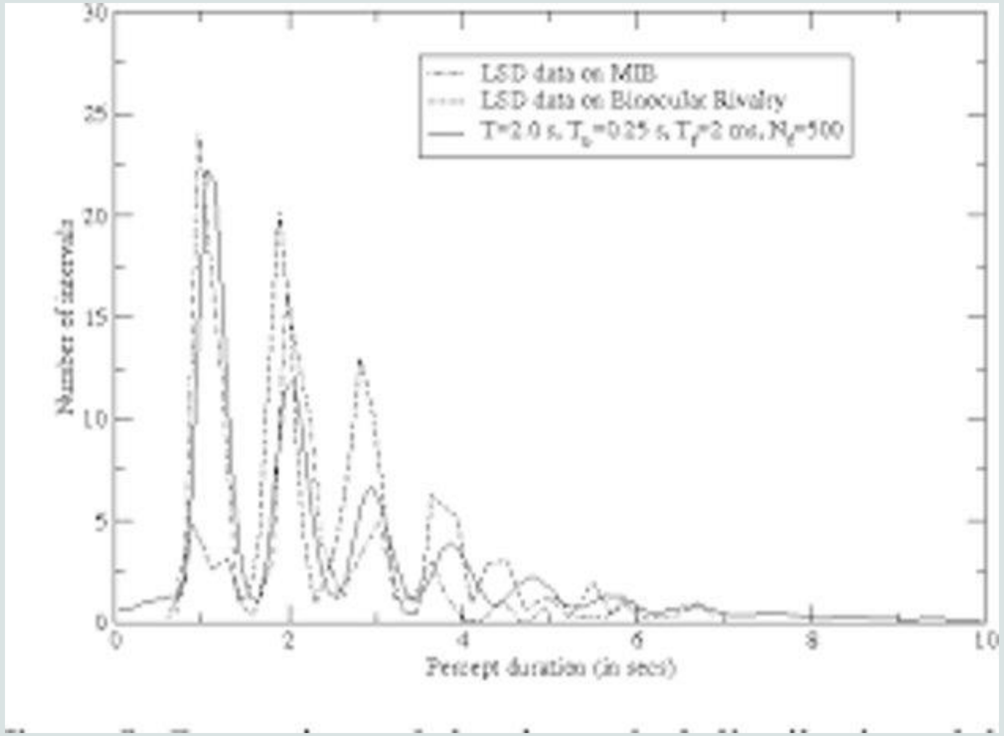
$$|c_1|^2 + |c_2|^2 = 1,$$

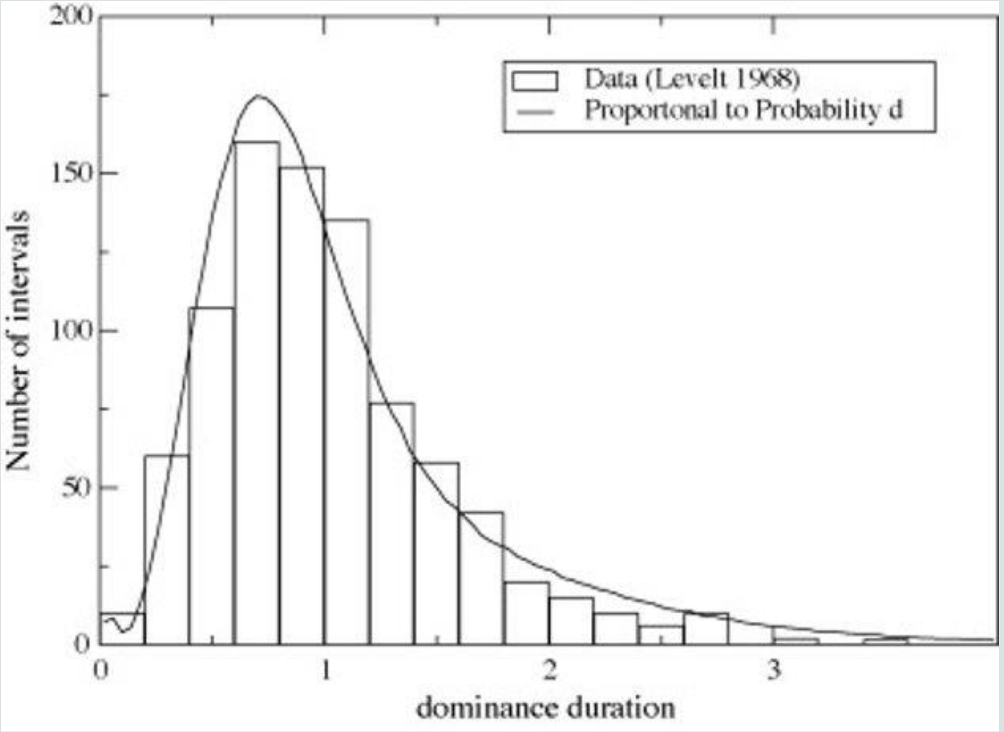
- **Measurement/projection in Binocular Rivalry:**Therefore, if we keep making conscious or involuntary attempts to observe frequently, the same state is projected (the quantum Zeno effect).
- Illustration of the neuronal activity or measurement events needed to understand the distribution of dominance duration in binocular rivalry



$$w_i = \sin^2(\bar{\omega}(t_i - t_{i-1})) \prod_{j=1}^{i-1} \cos^2(\bar{\omega}(t_j - t_{j-1})). \quad (22)$$

Result and comparison with Experiment





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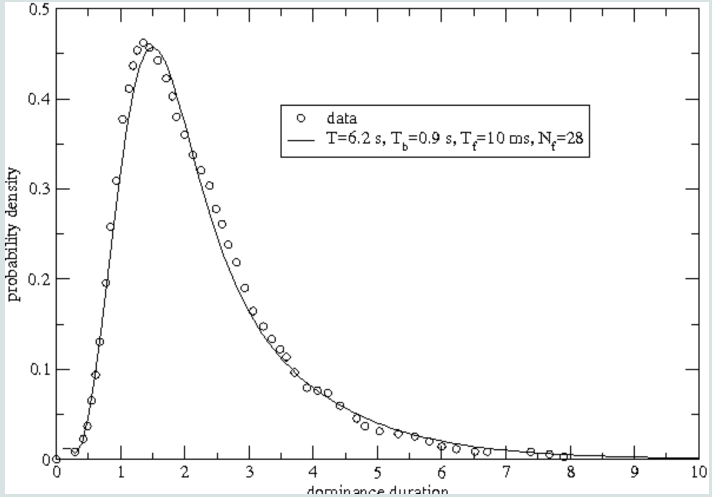
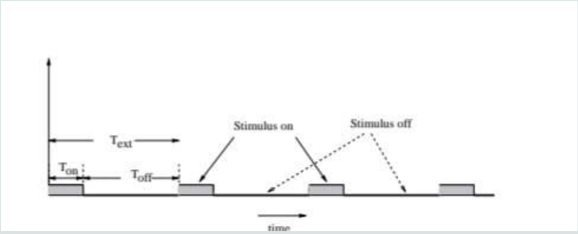


- **Probability distribution of dominance duration**: Neuron firing frequencies of the order of 100Hz have been used in these calculations.The firing patterns might be somewhat different for neurons of the human cerebral or extra striate cortex, which is the case of our interest; therefore, the values found, in order to achieve good agreement with the observed PDDD.
- **Periodic removal of stimulus:**The stimulus appears and disappears periodically and both rival images are presented to awareness for a time interval **Ton**, and for a time interval **Toff** both images are removed. It was found that the frequency of perceptual alterations can be slowed down, and even brought to almost a standstill. These experimental results are very important because they provide clear support for the present theory.



Discussion

- The present explanation of the phenomenon observed, when the stimulus was periodically removed, is based on the fact that our theory places the attention of consciousness higher, in the **hierarchy of consciousness**, than the two stimulated neural correlates in the brain.
- There are two types of factors which can enhance attention such as **Bottom-up factors:**If we make the diagonal terms of the operator ˆω different, the probability distribution of dominance durations for state |1i and for |2i would be equal. and **Top down factors** :Instructing the observer to pay attention to one particular perceptual state influences and modulates the frequency of measurements.

The issue of Decoherence

- If we consider the formalism of quantum theory as a mathematical foundation that describes the microscopic world, it is very difficult to make a convincing argument that the brain dynamics is governed by quantum theory because of **the issue of decoherence.**
- Therefore, in order to describe the external world as it is perceived by our conscious apparatus, we need to treat it using the same formulation used to describe the process of perception and thought. Otherwise we would use two different ways of describing our interaction with the external stimuli and this would be inconsistent.

Conclusion

- In summary, the present theory accurately describes:
- (a) the distribution of dominance duration in binocular rivalry.
- (b) the qualitative change in distribution of dominance duration and the appearance of oscillations in subjects which were studied under the influence of hallucinogenic drugs which, as is found, increase the neuron firing rates and decrease the potential frequency.
- (c) the marked increase in dominance duration by periodic removal of the stimulus. Furthermore, for binocular rivalry experiments where the stimulus is periodically removed, a distribution of perceptual alterations as a function of time is predicted and this can be tested in future experiments.

- Thank you